

$$V_X > V_Z > V_Y$$

- 21 C** p.d. across the $400\ \Omega$ resistor = p.d. across the $600\ \Omega$ resistor

Since $I = \frac{V}{R}$,

$$\text{Current through the } 600\ \Omega \text{ resistor} = \left(\frac{400}{400 + 600} \right) I = \frac{2}{5} I$$

$$\frac{\text{power dissipated across } 120\ \Omega}{\text{power dissipated across } 600\ \Omega} = \frac{I^2 (120)}{\left(\frac{2}{5} I \right)^2 (600)} = 1.25 = 1.3$$

- 22 D** Effective e.m.f. = 2.0 V